

Feasibility-Aware Scenario Learning for Carbon-Constrained Battery Planning in Data Centers

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Abstract

Planning battery storage for a data center requires a single design to balance annual electricity cost, outage service, and the carbon intensity of delivered electricity. The design can be determined by rare combinations of high demand, low photovoltaic availability, carbon-intensive grid supply, and grid outages; scenario supports selected for statistical similarity need not preserve these decision-critical events. This paper develops a two-stage battery-planning model and a feasibility-aware method for learning a compact scenario support. The planning model uses a source- and charging-time-resolved storage ledger that tracks initial, photovoltaic-, diesel-, and grid-charged energy without a bilinear perfect-mixing approximation. A conditional variational autoencoder is then fine-tuned using fixed-design recourse from the mixed-integer planning oracle. The decision loss combines an infeasibility-penalizing term, which corrects optimistic demand, photovoltaic, and carbon trajectories, with an optimality-preserving term, which prevents uniformly conservative scenarios from excluding the design preferred for matched real data. Solver outputs enter the loss only as detached violation severities and carbon-exposure weights; neither the optimizer nor decision regret is differentiated. The evaluation compares learned, statistical-selection, and predict-then-optimize supports using common held-out scenarios, installed capacity, annual cost, load shedding, carbon excess, and decision regret.

Keywords: Battery energy storage, carbon accounting, data center, decision-focused learning, scenario generation, stochastic planning

Nomenclature

Sets and Indices

$\mathcal{S}$	Planning scenarios, indexed by s .
$\mathcal{T}, \mathcal{T}^+$	Operating intervals and storage-state intervals, indexed by t .
$\mathcal{J}$	Supply sources {G, PV, DG}, indexed by j .
$\mathcal{K}$	Source- and vintage-resolved storage layers, indexed by k .
$\mathcal{I}$	Generated-support positions, indexed by n .
$\mathcal{D}^{\text{tr}}$	Historical training pool.
L, ES	Facility-load and energy-storage destinations.

Scenario and System Parameters

$D_{s,t}, P_{s,t}^{\text{PV,av}}$	Facility demand and available PV power.
$\lambda_{s,t}, \chi_{s,t}$	Grid energy price and carbon intensity.
$A_{s,t}$	Grid-availability indicator.
π_s, ν_s	Physical probability and annual occurrence factor.

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Δt	Duration of one operating interval.
$\eta^{\text{ch}}, \eta^{\text{dc}}$	Storage charging/discharging efficiencies.
$\sigma^{\text{sd}}, \kappa$	Self-discharge fraction and interval retention factor.
$\sigma^0, \underline{\sigma}, \overline{\sigma}$	Initial, minimum, and maximum state-of-charge fractions.
$\underline{P}^{\text{ES}}, \overline{P}^{\text{ES}}$	Minimum/maximum storage power if built.
$\underline{E}^{\text{ES}}, \overline{E}^{\text{ES}}$	Minimum/maximum storage energy if built.
$\underline{h}, \overline{h}$	Minimum/maximum storage duration.
$\overline{P}^{\text{G}}, \overline{P}^{\text{DG}}$	Grid-import and diesel-generator limits.
χ^0, χ^{DG}	Initial-inventory and diesel carbon intensities.
$\bar{\chi}^{\text{n}}, \bar{\chi}^0$	Normal-operation and outage carbon-intensity limits.
$\chi_{s,k}^{\text{ES}}$	Internal carbon intensity of storage layer k .
<i>Cost Parameters</i>	
CRF	Capital recovery factor.
$c^{\text{site}}, c^{\text{P}}, c^{\text{E}}$	Fixed, power-related, and energy-related storage investment costs.
$c^{\text{dem}}, c^{\text{DG}}$	Annual demand-charge rate and diesel energy cost.
$c^{\text{deg}}, c^{\text{cur}}, c^{\text{sh}}$	Storage degradation, PV-curtailment, and load-shedding costs.
c^{exc}	Penalty per unit of carbon-budget excess.
<i>Planning and Operating Variables</i>	
$u^{\text{ES}}, p^{\text{ES}}, E^{\text{ES}}$	Storage build, power-capacity, and energy-capacity decisions.
p^{pk}	Maximum grid import over the planning support.
$p_{s,t}^{j\text{L}}, p_{s,t}^{j\text{ES}}$	Power from source j to the load and storage.
$p_{s,t}^{\text{ch}}, p_{s,t}^{\text{dc}}$	Aggregate storage charging and discharging power.
$p_{s,t}^{\text{sh}}, p_{s,t}^{\text{cur}}$	Load shedding and PV curtailment.
$u_{s,t}^{\text{ch}}, u_{s,t}^{\text{dc}}$	Storage charge/discharge mode indicators.
$e_{s,k,t}, p_{s,k,t}^{\text{dc}}$	Energy and discharge power of layer k .
$\Delta e_{s,k,t}^{\text{in}}$	Internal energy added to layer k .
$m_{s,t}^{\text{L}}, \varepsilon_{s,t}^{\text{co2}}$	Carbon delivered to load and hourly carbon excess.

Learning Symbols

G_θ	Conditional scenario decoder.
$\widehat{\mathcal{S}}_\theta$	Scenario support generated by G_θ .
$\widehat{\mathbf{y}}_\theta, \mathbf{y}^\star$	Generated-support and matched-real reference designs.
$\delta^{\text{sh}}, \delta^{\text{co}2}$	Detached shedding and carbon-excess severities returned by recourse.
$H_{n,t,\tau}$	Detached exposure of delivered carbon in t to grid carbon intensity in τ .
F, P	Forward-feasibility and preservation-recourse superscripts.
$\epsilon_D, \epsilon_H, \epsilon_g, \epsilon_J$	Positive numerical safeguards.
$L_{\text{IPL}}, L_{\text{OPL}}$	Infeasibility-penalizing and optimality-preserving losses.
$\mathcal{R}_m$	Normalized out-of-sample decision regret of method m .

1. Introduction

Data centers combine large, persistent electricity demand with stringent service-continuity requirements. Behind-the-meter photovoltaic (PV) generation and battery storage can reduce demand charges, shift grid imports toward lower-carbon hours, and sustain critical demand during an outage. These services cannot be planned independently. A battery selected for price arbitrage may lack sufficient power or energy during an interruption, whereas a design chosen only for reliability may be charged in carbon-intensive hours. The planning problem must therefore represent economic performance, outage operation, and delivered carbon under the same uncertain trajectories.

Carbon-aware storage creates an accounting problem in addition to the physical energy balance. Grid electricity charged at different times can have different carbon intensities, and PV-, diesel-, and grid-charged electricity should not lose its identity merely because it occupies one physical battery. A perfectly mixed inventory assigns a single endogenous average intensity to all stored energy. Besides introducing bilinear relations, that abstraction discards the source and charging-time information that determines the carbon content of a later discharge. Research on carbon-aware power flow has shown why emissions must be attributed to delivered electricity [1]; storage extends the same attribution question across time.

Uncertainty creates a second difficulty. A planning scenario must jointly represent facility demand, PV availability, energy price, grid carbon intensity, and grid availability. Historical data provide many such trajectories, but repeatedly solving the resulting mixed-integer planning model on the complete pool is computationally burdensome. Classical scenario reduction approximates the empirical distribution under a statistical distance [2]. A statistically representative support, however, need not preserve the events that determine storage capacity. An error concentrated in an outage hour or in a carbon-intensive charging hour can change the design even when its contribution to average reconstruction error is small.

Decision-focused learning aligns prediction with downstream optimization [3], but the present problem does not admit a convenient end-to-end derivative. The planning oracle contains build and operating binaries, and only the first-stage battery design transfers from generated scenarios to realized operation. Moreover, generated demand, PV availability, and grid carbon intensity appear in feasibility constraints. An optimistic support can produce an inexpensive design that sheds load or exceeds its carbon budget under real recourse; a uniformly conservative support can avoid these violations while excluding an otherwise effective design. This distinction motivates the feasibility/optimality-preservation framework in [4].

This paper combines exact storage planning with conditional scenario generation. A conditional variational autoencoder (CVAE) [5] produces a fixed support of 48-hour trajectories. The planning oracle selects a battery from that support, after which the design is fixed and every operating variable is reoptimized on matched real trajectories. Detached load-shedding, carbon-excess, and carbon-exposure outputs identify optimistic generated constraints. In the reverse direction, a design optimized for the matched real batch is fixed and re-dispatched on the generated support to identify excessive conservatism.

The main contributions are as follows.

1. A two-stage data-center battery-planning model jointly represents investment, energy and demand charges, outage-only load shedding, and hourly load-side carbon limits. A linear source- and charging-time-resolved ledger preserves the carbon intensity of stored electricity without the bilinear identity required by a perfectly mixed inventory.
2. A feasibility-aware scenario-learning method is developed for mixed-integer, two-stage planning. Fixed-design recourse generates infeasibility-penalizing and optimality-preserving signals, while detached solver outputs avoid differentiation through the planning oracle.
3. A controlled evaluation protocol uses disjoint data splits, identical support sizes, common held-out recourse, and a perfect-information reference to compare annual cost, physical violations, installed capacity, storage value, and decision regret.

Section 2 formulates the storage-planning problem. Section 3 presents the decision-focused methodology. Section 4 describes the case study, and Section 5 concludes the paper.

Conventional decision-focused learning primarily aligns predictions with the quality of the resulting optimization objective. When the learned quantities parameterize constraints, however, objective quality alone is insufficient. We further require the design induced by the generated scenarios to remain feasible under realized operation, without making the learned support so conservative that it excludes high-quality designs.

2. Carbon-Aware Storage Planning Formulation

2.1. Problem Definition and System Components

Consider a *carbon-aware storage planning* (CASP) problem for a grid-connected data center subject to an operational carbon-intensity limit. The CASP is formulated as a two-stage stochastic program. The first stage optimizes the storage investment, i.e., the installation decision and its power and energy ratings. For each scenario $s \in \mathcal{S} = \{1, \dots, S\}$, the second stage optimizes system operation over the operating horizon $\mathcal{T} = \{1, \dots, T\}$, subject to load-service and carbon-intensity constraints.

The system comprises an aggregate data-center load (L), the utility grid (G), behind-the-meter photovoltaic generation (PV), a dispatchable diesel generator (DG), and a candidate battery energy storage system (ES). The supply sources are indexed by $\mathcal{J} = \{G, PV, DG\}$ and can serve the facility load directly or charge the storage system. Storage states are indexed by $\mathcal{T}^+ = \{1, \dots, T + 1\}$.

2.2. Planning Objective

The first-stage design is represented by

$$\mathbf{y} = (u^{\text{ES}}, P^{\text{ES}}, E^{\text{ES}}), \quad (1)$$

where u^{ES} indicates whether storage is installed and P^{ES} and E^{ES} are its rated power and energy. Its annualized investment cost is

$$C^{\text{inv}}(\mathbf{y}) = \text{CRF} \left(c^{\text{site}} u^{\text{ES}} + c^P P^{\text{ES}} + c^E E^{\text{ES}} \right). \quad (2)$$

Scenario $s \in \mathcal{S}$ is the joint trajectory

$$\xi_s = \{D_{s,t}, P_{s,t}^{\text{PV,av}}, \lambda_{s,t}, \chi_{s,t}, A_{s,t}\}_{t \in \mathcal{T}}, \quad (3)$$

where $D_{s,t}$ is demand, $P_{s,t}^{\text{PV,av}}$ is available PV power, $\lambda_{s,t}$ is the grid energy price, $\chi_{s,t}$ is the grid carbon intensity, and $A_{s,t} \in \{0, 1\}$ is the grid-availability indicator. If π_s is the physical probability of scenario s , its annual occurrence factor is

$$\nu_s = \pi_s \frac{8760}{T \Delta t}, \quad (4)$$

where $\pi_s \geq 0$ and $\sum_{s \in \mathcal{S}} \pi_s = 1$. Importance-corrected probabilities prevent oversampled outage scenarios from inflating their assumed annual frequency.

Let p^{pk} be the maximum total grid import over all scenarios. The annual operating cost is

$$\begin{aligned} C^{\text{op}} = & c^{\text{dem}} p^{\text{pk}} + \sum_{s \in \mathcal{S}} \nu_s \Delta t \sum_{t \in \mathcal{T}} [\lambda_{s,t} (p_{s,t}^{\text{G,L}} + p_{s,t}^{\text{G,ES}}) \\ & + c^{\text{DG}} (p_{s,t}^{\text{DG,L}} + p_{s,t}^{\text{DG,ES}}) + c^{\text{deg}} (p_{s,t}^{\text{ch}} + p_{s,t}^{\text{dc}})] \end{aligned}$$

$$+ c^{\text{cur}} p_{s,t}^{\text{cur}} + c^{\text{sh}} p_{s,t}^{\text{sh}}], \quad (5)$$

Carbon excess, when enabled during training and diagnosis, incurs

$$C^{\text{exc}} = c^{\text{exc}} \sum_{s \in \mathcal{S}} \nu_s \Delta t \sum_{t \in \mathcal{T}} \varepsilon_{s,t}^{\text{co2}}. \quad (6)$$

For the horizon-average benchmark, this term becomes $C^{\text{exc}} = c^{\text{exc}} \sum_{s \in \mathcal{S}} \nu_s \varepsilon_s^{\text{co2}}$. The planning oracle minimizes

$$C^{\text{ann}} = C^{\text{inv}} + C^{\text{op}} + C^{\text{exc}}. \quad (7)$$

When a hard carbon limit is required, $\varepsilon_{s,t}^{\text{co2}}$ is fixed to zero and C^{exc} vanishes. Soft-cap solves retain this variable so carbon infeasibility can be penalized and compared on a common recourse problem.

2.3. Storage Design and Site Dispatch

The build decision links the installed capacities to their admissible ranges:

$$\underline{P}^{\text{ES}} u^{\text{ES}} \leq P^{\text{ES}} \leq \overline{P}^{\text{ES}} u^{\text{ES}}, \quad (8)$$

$$\underline{E}^{\text{ES}} u^{\text{ES}} \leq E^{\text{ES}} \leq \overline{E}^{\text{ES}} u^{\text{ES}}, \quad (9)$$

$$\underline{h} P^{\text{ES}} \leq E^{\text{ES}} \leq \overline{h} P^{\text{ES}}. \quad (10)$$

Here, $u^{\text{ES}} \in \{0, 1\}$ and $p^{\text{pk}} \geq 0$. Thus, u^{ES} determines installation rather than location.

For source $j \in \mathcal{J}$, $p_{s,t}^{j,\text{L}}$ supplies the facility directly and $p_{s,t}^{j,\text{ES}}$ charges storage. The facility and PV balances are

$$\sum_{j \in \mathcal{J}} p_{s,t}^{j,\text{L}} + p_{s,t}^{\text{dc}} = D_{s,t} - p_{s,t}^{\text{sh}}, \quad (11)$$

$$p_{s,t}^{\text{PV,L}} + p_{s,t}^{\text{PV,ES}} + p_{s,t}^{\text{cur}} = p_{s,t}^{\text{PV,av}}. \quad (12)$$

Grid availability, generator rating, and the demand-charge epigraph require

$$p_{s,t}^{\text{G,L}} + p_{s,t}^{\text{G,ES}} \leq A_{s,t} \overline{P}^{\text{G}}, \quad (13)$$

$$p_{s,t}^{\text{DG,L}} + p_{s,t}^{\text{DG,ES}} \leq \overline{P}^{\text{DG}}, \quad (14)$$

$$p_{s,t}^{\text{G,L}} + p_{s,t}^{\text{G,ES}} \leq p^{\text{pk}}. \quad (15)$$

Load shedding is restricted to outages:

$$0 \leq p_{s,t}^{\text{sh}} \leq D_{s,t} (1 - A_{s,t}). \quad (16)$$

Consequently, a carbon limit cannot be met by dropping grid-connected load.

Aggregate charging and discharging are

$$p_{s,t}^{\text{ch}} = \sum_{j \in \mathcal{J}} p_{s,t}^{j,\text{ES}}, \quad (17)$$

$$p_{s,t}^{\text{dc}} = \sum_{k \in \mathcal{K}} p_{s,t}^{\text{dc},k}. \quad (18)$$

Binary variables $u_{s,t}^{\text{ch}}$ and $u_{s,t}^{\text{dc}}$ prevent simultaneous charging and discharging:

$$p_{s,t}^{\text{ch}} \leq P^{\text{ES}}, \quad p_{s,t}^{\text{ch}} \leq \overline{P}^{\text{ES}} u_{s,t}^{\text{ch}}, \quad (19)$$

$$p_{s,t}^{\text{dc}} \leq P^{\text{ES}}, \quad p_{s,t}^{\text{dc}} \leq \overline{P}^{\text{ES}} u_{s,t}^{\text{dc}}, \quad (20)$$

$$u_{s,t}^{\text{ch}} + u_{s,t}^{\text{dc}} \leq u^{\text{ES}}. \quad (21)$$

The mode variables are binary, and all power-flow variables are nonnegative.

2.4. Source- and Vintage-Resolved Carbon Accounting

To retain the origin of stored electricity, the battery inventory is partitioned into the virtual layers

$$\mathcal{K} = \{0, \text{PV}, \text{DG}\} \cup \{G_\tau : \tau \in \mathcal{T}\}, \quad (22)$$

where 0, PV, and DG denote initial, PV-charged, and diesel-charged inventory, while G_τ denotes grid energy charged in interval τ . These layers share the same physical power and energy ratings; they are carbon bookkeeping states, not separate battery compartments.

The internal carbon intensity assigned to each layer is

$$\chi_{s,k}^{\text{ES}} = \begin{cases} \chi^0, & k = 0, \\ 0, & k = \text{PV}, \\ \chi^{\text{DG}}/\eta^{\text{ch}}, & k = \text{DG}, \\ \chi_{s,\tau}/\eta^{\text{ch}}, & k = G_\tau. \end{cases} \quad (23)$$

The division by η^{ch} expresses source emissions per unit of internal stored energy: charging losses reduce energy entering the cell but do not remove emissions already attributed to that electricity.

The energy added to layer k in interval t is

$$\Delta e_{s,k,t}^{\text{in}} = \begin{cases} \eta^{\text{ch}} p_{s,t}^{\text{PV,ES}} \Delta t, & k = \text{PV}, \\ \eta^{\text{ch}} p_{s,t}^{\text{DG,ES}} \Delta t, & k = \text{DG}, \\ \eta^{\text{ch}} p_{s,t}^{\text{G,ES}} \Delta t, & k = G_t, \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

Let $\kappa = 1 - \sigma^{\text{sd}}$ be the one-interval retention factor. For $s \in \mathcal{S}$, $k \in \mathcal{K}$, and $t \in \mathcal{T}$, layer energy evolves according to

$$e_{s,k,t+1} = \kappa e_{s,k,t} + \Delta e_{s,k,t}^{\text{in}} - \frac{p_{s,k,t}^{\text{dc}}}{\eta^{\text{dc}}} \Delta t, \quad (25)$$

$$\frac{p_{s,k,t}^{\text{dc}}}{\eta^{\text{dc}}} \Delta t \leq \kappa e_{s,k,t}. \quad (26)$$

The initial layer allocation and aggregate state-of-charge bounds are

$$e_{s,0,1} = \sigma^0 E^{\text{ES}}, \quad e_{s,k,1} = 0, \quad k \neq 0, \quad (27)$$

$$\underline{\sigma} E^{\text{ES}} \leq \sum_{k \in \mathcal{K}} e_{s,k,t} \leq \overline{\sigma} E^{\text{ES}}, \quad t \in \mathcal{T}^+. \quad (28)$$

Each representative block returns to its initial stored energy and cannot end with a larger carbon inventory:

$$\sum_{k \in \mathcal{K}} e_{s,k,T+1} = \sigma^0 E^{\text{ES}}, \quad (29)$$

$$\sum_{k \in \mathcal{K}} \chi_{s,k}^{\text{ES}} e_{s,k,T+1} \leq \chi^0 \sigma^0 E^{\text{ES}}. \quad (30)$$

The carbon-delivery rate assigned to served facility load is

$$m_{s,t}^{\text{L}} = \chi_{s,t} p_{s,t}^{\text{G,L}} + \chi^{\text{DG}} p_{s,t}^{\text{DG,L}} + \sum_{k \in \mathcal{K}} \chi_{s,k}^{\text{ES}} \frac{p_{s,k,t}^{\text{dc}}}{\eta^{\text{dc}}}. \quad (31)$$

Carbon associated with charging enters the inventory ledger and is attributed to the facility only when the stored energy is discharged. Hence, $m_{s,t}^{\text{L}}$ is a load-side compliance rate; it is not the source-side emissions rate, which also counts grid and diesel energy used for charging.

2.5. Carbon Limits and Benchmark Accounting

The carbon-intensity limit depends on grid availability:

$$\bar{\chi}_{s,t} = A_{s,t}\bar{\chi}^n + (1 - A_{s,t})\bar{\chi}^o. \quad (32)$$

The principal formulation imposes the following hourly load-side budget:

$$m_{s,t}^L \leq \bar{\chi}_{s,t}(D_{s,t} - p_{s,t}^{\text{sh}}) + \varepsilon_{s,t}^{\text{co2}}, \quad \varepsilon_{s,t}^{\text{co2}} \geq 0. \quad (33)$$

For comparison, a horizon-average cap replaces (33) with

$$\sum_{t \in \mathcal{T}} m_{s,t}^L \Delta t \leq \sum_{t \in \mathcal{T}} \bar{\chi}_{s,t}(D_{s,t} - p_{s,t}^{\text{sh}}) \Delta t + \varepsilon_s^{\text{co2}}. \quad (34)$$

The layered formulation is linear apart from the build and operating-mode binaries. A perfectly mixed benchmark instead tracks energy $E_{s,t}^{\text{mix}}$, carbon inventory $M_{s,t}^{\text{mix}}$, and average internal intensity $\chi_{s,t}^{\text{mix}}$ through

$$M_{s,t}^{\text{mix}} = \chi_{s,t}^{\text{mix}} E_{s,t}^{\text{mix}}, \quad m_{s,t}^{B,\text{dc}} = \chi_{s,t}^{\text{mix}} \frac{p_{s,t}^{\text{dc}}}{\eta^{\text{dc}}}. \quad (35)$$

These bilinear identities make the mixed-inventory benchmark nonconvex. The benchmark is used only to determine whether discarding source and charging-time information changes the resulting storage design.

2.6. Compact Formulation

Let $\mathbf{x}_s$ collect the recourse variables for scenario s , let $\mathcal{Y}$ denote the design set in (8)–(10), and let $\mathcal{X}(\mathbf{y}, \xi_s)$ denote the feasible operating set defined by (11)–(33). The deterministic equivalent is

$$\begin{aligned} \min_{\mathbf{y}, p^{\text{pk}}, \{\mathbf{x}_s\}} \quad & C^{\text{ann}}(\mathbf{y}, p^{\text{pk}}, \{\mathbf{x}_s\}) \\ \text{s.t.} \quad & \mathbf{y} \in \mathcal{Y}, \\ & \mathbf{x}_s \in \mathcal{X}(\mathbf{y}, \xi_s), \quad s \in \mathcal{S}. \end{aligned} \quad (36)$$

Equation (36) is the deterministic equivalent of CASP and serves as the planning oracle throughout the learning procedure.

3. Decision-Focused Learning Methodology

3.1. Learning Task and Scenario Generator

Consider a historical pool $\mathcal{D}^{\text{tr}}$ whose direct use in every planning solve is computationally prohibitive. For a support budget $K \ll |\mathcal{D}^{\text{tr}}|$, define $\mathcal{I} = \{1, \dots, K\}$. The learning task is to generate K trajectories that retain the planning information in the full pool rather than merely its average statistical shape. The conditional decoder produces

$$\widehat{\mathcal{S}}_\theta = \{\widehat{\xi}_n = G_\theta(\mathbf{z}_n, \omega_n) : n \in \mathcal{I}\}, \quad (37)$$

where $\mathbf{z}_n$ is a latent draw and ω_n contains cyclical day-of-year coordinates, weekend share, and mean temperature. The contexts, latent draws, outage indicators, and support weights are held fixed during decision-focused fine-tuning. Consequently, a change in the induced design can be attributed to an update of G_θ , not to resampling. Support weights preserve the importance-corrected probability mass of normal and outage cases.

The CVAE encoder is

$$q_\phi(\mathbf{z} \mid \mathbf{x}, \omega) = \mathcal{N}(\boldsymbol{\mu}_\phi, \text{diag}(\boldsymbol{\sigma}_\phi^2)), \quad (38)$$

and the statistical training loss is

$$\begin{aligned} L_{\text{stat}} = & L_{\text{rec}} + w_{\text{ramp}} L_{\text{ramp}} + w_{\text{net}} L_{\text{net}} + w_{\text{pk}} L_{\text{pk}} \\ & + w_{\lambda} L_{\lambda\text{-spread}} + w_{\chi} L_{\chi\text{-spread}} + \beta L_{\text{KL}}. \end{aligned} \quad (39)$$

Here, L_{rec} is a field-weighted reconstruction loss; L_{ramp} preserves hourly changes; and the remaining shape terms preserve net load, peak net load, price spread, and carbon-intensity spread. Only D , $P^{\text{PV,av}}$, and χ carry gradients from the decision loss. Workload and the calendar tariff are restored from admissible templates, while A is copied as a detached exogenous contingency. The generator therefore cannot create arbitrage prices or alter outage occurrence to improve its loss.

3.2. Forward Planning and Fixed-Design Recourse

Solving the planning oracle on the generated support yields

$$\widehat{\mathbf{y}}_\theta \in \operatorname{argmin}_{\mathbf{y} \in \mathcal{Y}} C^{\text{ann}}(\mathbf{y}; \widehat{\mathbf{S}}_\theta). \quad (40)$$

Only the first-stage design $\widehat{\mathbf{y}}_\theta$ is transferred to the matched real trajectory ξ_n . Its operating variables are discarded and recourse is solved again:

$$\mathbf{x}_n^F \in \operatorname{argmin}_{\mathbf{x} \in \mathcal{X}(\widehat{\mathbf{y}}_\theta, \xi_n)} C^{\text{rec}}(\mathbf{x}; \widehat{\mathbf{y}}_\theta, \xi_n). \quad (41)$$

All dispatch, mode, inventory, curtailment, shedding, and carbon variables are therefore adapted to the realized trajectory. Let $\delta_{n,t}^{\text{sh},F}$ and $\delta_{n,t}^{\text{co2},F}$ denote the hourly load shedding and carbon excess returned by (41), and let $H_{n,t,\tau}^F$ denote the corresponding exposure of delivered carbon in hour t to grid carbon intensity in hour τ . Solver decisions, violation severities, and exposures are detached before the loss is formed. They weight decision-critical prediction errors but do not define a derivative through the mixed-integer optimizer.

3.3. Infeasibility-Penalizing Loss

The infeasibility-penalizing loss (IPL) corrects optimistic generated constraints only when the induced design fails under real recourse. Define the normalization terms

$$b_{n,t}^D = \max(|D_{n,t}|, \epsilon_D), \quad b_{n,t}^{\text{srV}} = \max(D_{n,t} - \delta_{n,t}^{\text{sh},F}, \epsilon_D), \quad (42)$$

and the directional residuals for demand and PV:

$$r_{n,t}^D = \frac{[D_{n,t} - \widehat{D}_{n,t}]_+}{b_{n,t}^D}, \quad r_{n,t}^{\text{PV}} = \frac{[\widehat{P}_{n,t}^{\text{PV,av}} - P_{n,t}^{\text{PV,av}}]_+}{b_{n,t}^D}. \quad (43)$$

These residuals are positive when generated demand is too low or generated PV availability is too high.

Carbon errors require an intertemporal direction because grid energy charged in hour τ may be discharged in hour t . To preserve a gradient near the normal-operation carbon limit, define the smooth carbon pressure

$$\varphi(\chi) = s_\chi \log \left[1 + \exp \left(\frac{\chi - \bar{\chi}^n}{s_\chi} \right) \right], \quad s_\chi > 0. \quad (44)$$

The exposure-weighted optimistic carbon residual is

$$r_{n,t}^\chi = \frac{[\sum_{\tau \in \mathcal{T}} H_{n,t,\tau}^F [\varphi(\chi_{n,\tau}) - \varphi(\bar{\chi}_{n,\tau})]]_+}{\bar{\chi}^n \max(\sum_{\tau \in \mathcal{T}} H_{n,t,\tau}^F, \epsilon_H)}. \quad (45)$$

For the zero-anchored smooth margin $h_m(r) = \text{softplus}(m+r) - \text{softplus}(m)$, the three IPL components are

$$L_D^{\text{IPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{sh},F}}{b_{n,t}^D} h_m(r_{n,t}^D), \quad (46)$$

$$L_{\text{PV}}^{\text{IPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{sh},F}}{b_{n,t}^D} h_m(r_{n,t}^{\text{PV}}), \quad (47)$$

$$L_\chi^{\text{IPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{co2},F}}{b_{n,t}^{\text{srV}}} h_m(r_{n,t}^\chi), \quad (48)$$

and

$$L_{\text{IPL}} = w_D L_D^{\text{IPL}} + w_{\text{PV}} L_{\text{PV}}^{\text{IPL}} + w_\chi L_\chi^{\text{IPL}}. \quad (49)$$

Gradient descent therefore raises underestimated demand, lowers overestimated PV availability, and raises underestimated carbon pressure only at locations weighted by real recourse violations.

3.4. Optimality-Preserving Loss

IPL alone can reduce violations by making every generated scenario severe. To prevent this degenerate solution, let $\mathcal{B}^R = \{\xi_n : n \in \mathcal{I}\}$ be the matched real batch and solve

$$\mathbf{y}^\star \in \underset{\mathbf{y} \in \mathcal{Y}}{\operatorname{argmin}} C^{\text{ann}}(\mathbf{y}; \mathcal{B}^R). \quad (50)$$

The reference design $\mathbf{y}^\star$ is fixed and re-dispatched on $\widehat{\mathcal{S}}_\theta$. Denote the resulting shedding, carbon excess, and carbon-exposure matrix by $\delta_{n,t}^{\text{sh},P}$, $\delta_{n,t}^{\text{co2},P}$, and $H_{n,t,\tau}^P$, respectively. A positive violation means that the generated constraints have excluded the design optimized for the matched real batch.

The OPL uses the reverse directional residuals

$$\bar{r}_{n,t}^D = \frac{[\widehat{D}_{n,t} - D_{n,t}]_+}{b_{n,t}^D}, \quad \bar{r}_{n,t}^{\text{PV}} = \frac{[P_{n,t}^{\text{PV,av}} - \widehat{P}_{n,t}^{\text{PV,av}}]_+}{b_{n,t}^D}, \quad (51)$$

and the exposure-weighted carbon residual

$$\bar{r}_{n,t}^\chi = \frac{\left[\sum_{\tau \in \mathcal{T}} H_{n,t,\tau}^P [\varphi(\widehat{\chi}_{n,\tau}) - \varphi(\chi_{n,\tau})] \right]_+}{\bar{\chi}^n \max(\sum_{\tau \in \mathcal{T}} H_{n,t,\tau}^P, \epsilon_H)}. \quad (52)$$

With $\widetilde{b}_{n,t}^{\text{srv}} = \max(\widehat{D}_{n,t} - \delta_{n,t}^{\text{sh},P}, \epsilon_D)$, define

$$L_D^{\text{OPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{sh},P}}{b_{n,t}^D} h_m(\bar{r}_{n,t}^D), \quad (53)$$

$$L_{\text{PV}}^{\text{OPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{sh},P}}{b_{n,t}^D} h_m(\bar{r}_{n,t}^{\text{PV}}), \quad (54)$$

$$L_\chi^{\text{OPL}} = \frac{1}{KT} \sum_{n \in \mathcal{I}} \sum_{t \in \mathcal{T}} \frac{\delta_{n,t}^{\text{co2},P}}{\widetilde{b}_{n,t}^{\text{srv}}} h_m(\bar{r}_{n,t}^\chi). \quad (55)$$

The complete preservation loss is

$$L_{\text{OPL}} = w_D L_D^{\text{OPL}} + w_{\text{PV}} L_{\text{PV}}^{\text{OPL}} + w_\chi L_\chi^{\text{OPL}}. \quad (56)$$

It lowers excessively high demand and carbon predictions and raises excessively low PV availability only where the generated support rejects the reference design.

Finally, the two directional losses are combined as

$$L_{\text{dec}} = \alpha L_{\text{IPL}} + (1 - \alpha) L_{\text{OPL}}, \quad 0 \leq \alpha \leq 1. \quad (57)$$

This construction adapts the feasibility/optimality logic of [4] to two-stage fixed-design recourse. It is a directional surrogate, not a derivative of the MILP value function.

3.5. Gradient Balancing and Training

Decision feedback and statistical reconstruction are evaluated on independent batches. Their combined objective is

$$L_{\text{total}} = w_{\text{stat}} L_{\text{stat}} + w_{\text{dec}}^{\text{eff}} L_{\text{dec}}. \quad (58)$$

Let g_{stat} and g_{dec} be the corresponding decoder gradient norms. The detached decision-loss weight is

$$w_{\text{dec}}^{\text{eff}} = w_{\text{dec}} \operatorname{clip} \left(r_g \frac{g_{\text{stat}}}{\max(g_{\text{dec}}, \epsilon_g)}, b_{\min}, b_{\max} \right). \quad (59)$$

Only decoder parameters are used in this comparison because both loss terms reach that parameter subspace, whereas only the statistical loss reaches the encoder.

Algorithm 1 Feasibility-aware CVAE fine-tuning

Require: CVAE, training and validation pools, planning oracle, support size K

- 1: Fix support contexts, latent draws, outage indicators, and weights
 - 2: **for** each fine-tuning epoch **do**
 - 3: Decode $\widehat{\mathcal{S}}_\theta$ and solve (40)
 - 4: Re-dispatch $\widehat{\mathbf{y}}_\theta$ on $\mathcal{B}^R$
 - 5: Solve (50) and re-dispatch $\mathbf{y}^\star$ on $\widehat{\mathcal{S}}_\theta$
 - 6: Form L_{IPL} , L_{OPL} , and L_{dec}
 - 7: Evaluate L_{stat} on an independent reconstruction batch
 - 8: Balance decoder gradients and update θ
 - 9: Evaluate the fixed support on the validation pool
 - 10: Restore the best validation checkpoint and return its K scenarios
-

Checkpoint selection is lexicographic. Checkpoints that exceed the prescribed shedding tolerance are rejected; among reliable checkpoints, lower carbon excess is preferred and exact decision regret breaks ties. The initial model is retained as an epoch-1 candidate, so decision-focused updates are accepted only when they improve the fixed validation criterion.

3.6. Out-of-Sample Metrics

Let $\mathcal{S}^{\text{te}}$ be the common held-out test set. Its perfect-information cost and the fixed-design cost of method m are

$$J^{\text{PI}} = \min_{\mathbf{y} \in \mathcal{Y}} C^{\text{ann}}(\mathbf{y}; \mathcal{S}^{\text{te}}), \quad (60)$$

$$J_m = C^{\text{ann}}(\mathbf{y}_m; \mathcal{S}^{\text{te}}), \quad (61)$$

$$\mathcal{R}_m = \frac{J_m - J^{\text{PI}}}{|J^{\text{PI}}| + \epsilon_J}. \quad (62)$$

The regret $\mathcal{R}_m$ is an evaluation metric and never enters gradient calculation. With J^0 denoting the no-storage cost on the same test set, the storage value is $\mathcal{V}_m = J^0 - J_m$. Annual cost is reported together with load shedding, carbon excess, PV curtailment, and the installed power and energy ratings.

4. Case Study

4.1. Data and Experimental Setup

The dataset is divided into nonoverlapping 48-hour trajectories covering three years. Facility demand is calibrated to a 10-MW-class data center using a workload proxy and temperature-dependent power-usage effectiveness. The PV shape is retained from the source feeder location and rescaled to a 5-MW installation. Electricity tariffs follow their calendar templates, and each grid-carbon trajectory remains an intact 48-hour block. Any within-season permutation is performed separately inside the training, validation, and test splits, which remain mutually disjoint.

Contiguous outages of one to four hours are injected into a stratified subset of trajectories so that small decision batches contain reliability events. Because outages are deliberately oversampled for learning, importance weights restore an assumed physical probability of approximately 0.71% per 48-hour window. Outage time is sampled independently of demand, PV, and carbon; it is therefore a controlled stress-test condition rather than an estimate of their joint field distribution.

The CVAE has a 16-dimensional latent space and two 256-unit hidden layers in both encoder and decoder. Statistical pretraining runs for 300 epochs and decision-focused fine-tuning for 45 epochs. The principal configuration fixes eight decision anchors, uses an independent 32-trajectory reconstruction batch, sets $\alpha = 0.5$, and targets a decoder-gradient ratio of $r_g = 0.1$. Planning solves used during training allow a 1% relative gap; final evaluation uses the tighter oracle tolerance.

Table 1: Principal system and planning parameters

Parameter	Value
Scenario horizon/resolution	48 h/1 h
Mean/maximum demand	8,477/10,224 MW
PV capacity	5 MW
Grid-import limit	15 MW
Backup-generator capacity	6 MW
Battery power bound if built	0.2–7.5 MW
Battery energy bound if built	0.5–60 MWh
Initial/minimum/maximum SOC	0.50/0.10/0.90
Charge/discharge efficiency	0.95/0.95
Normal/outage carbon limit	0.28/0.66 tCO ₂ /MWh
Planning support size	8

Table 2: Out-of-sample planning performance on a common test set

Method	P^{ES} (MW)	E^{ES} (MWh)	Annual cost (\$)	$\mathcal{R}$ (%)	Shedding (MWh)	Carbon excess (tCO ₂)	Storage value (\$)
Random	—	—	—	—	—	—	—
K-means	—	—	—	—	—	—	—
Farthest	—	—	—	—	—	—	—
Aggregate	—	—	—	—	—	—	—
CVAE-only	—	—	—	—	—	—	—
IPL-only	—	—	—	—	—	—	—
OPL-only	—	—	—	—	—	—	—
Full DFL	—	—	—	—	—	—	—
Perfect information	—	—	—	0	—	—	—

4.2. Compared Methods and Metrics

The statistical-selection baselines are random sampling, K-means, farthest-point selection, and an aggregate observed support. Learning baselines include a statistically trained CVAE, IPL-only fine-tuning, and OPL-only fine-tuning; the proposed method uses both directional losses. Every method receives the same support budget and is evaluated by fixing its selected design and solving recourse on the identical test trajectories. Direct optimization on the full test set is used only to define the perfect-information reference.

The completed analysis will first compare the selected power and energy ratings, then determine whether cost differences are accompanied by load shedding or carbon excess, and finally compare the observed differences with both multi-seed variability and the uncertainty implied by solver gaps.

4.3. Carbon-Accounting and Learning Ablations

Four matched planning models compare source-resolved and perfectly mixed carbon accounting under hourly and horizon-average carbon limits. Data, scenario weights, cost coefficients, design bounds, and solver settings are held fixed across all four models.

The learning ablation varies α , removes decoder-gradient balancing, and changes the support size. Planning sensitivities vary the normal-operation carbon limit, battery capital cost, and outage duration. Multi-seed results are reported as means and standard deviations. A representative 48-hour dispatch will show grid and delivered carbon intensity, aggregate state of charge, charging source, and discharge by carbon layer so that differences in selected capacity can be traced to operating behavior.

4.4. Computational Performance and Limitations

Computational reporting separates forward-planning, fixed-design recourse, and matched-real reference solves; it also records solver calls per epoch, wall time, and final optimality gaps. The aggregate facility model cannot support

Table 3: Effect of carbon accounting and cap scope

Accounting	Scope	P^{ES} (MW)	E^{ES} (MWh)	Excess (tCO ₂)
Layered	Hourly	–	–	–
Layered	Horizon	–	–	–
Average	Hourly	–	–	–
Average	Horizon	–	–	–

claims about network congestion or multi-location siting. Demand is calibrated rather than metered, outage time is simulated independently of weather, and the directional decision loss is not a derivative of the MILP value function.

5. Conclusion

This paper has developed a two-stage formulation for carbon-constrained battery planning in a grid-connected data center. The model jointly accounts for investment, energy purchases, demand charges, storage degradation, outage-only load shedding, and the carbon intensity delivered to the facility. Virtual inventory layers retain both energy source and grid-charging time while sharing the physical power and energy limits of one battery. The resulting ledger is linear and avoids the bilinear intensity identities of a perfectly mixed carbon inventory.

A feasibility-aware learning method has also been developed to construct a compact planning support. In the forward direction, the design selected from generated scenarios is fixed and re-dispatched on matched real trajectories; in the preservation direction, a matched-real reference design is fixed and re-dispatched on generated trajectories. The corresponding shedding, carbon excess, and carbon-exposure outputs distinguish optimism from excessive conservatism. Because these solver outputs are detached, the method retains gradients to generated demand, PV availability, and carbon intensity without differentiating through the mixed-integer oracle.

Future work will consider network-constrained multi-location planning, weather-correlated outages, flexible computing demand, and longer horizons with linked inter-block storage states.

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CRedit authorship contribution statement

Author Name: Conceptualization, Methodology, Software, Validation, Writing—original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed data, configurations, and source code used to reproduce the study will be made available with the published article.

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